DAPHNE–SSP scan report

Editable operator template

[Operator and institute]

[Date, time range, and time zone]

Instructions. Replace bracketed fields with observations and artifact paths. Delete instructions before circulation. Mark a check as failed or not tested when evidence is absent. The reference run 45371 belongs to a separate completed scan; do not reuse its numbers as a new measurement.

1 Purpose and result

[One paragraph: what was scanned, which channels were acquired, whether it completed, and the principal finding. Distinguish an electronics/control exercise from a detector gain measurement.]

Identity	Recorded value
Board / location	*[Asset ID; rack; installation or cabling changes]*
Operator / control host	*[Authorized operating account and host]*
DAQ run / scan tag	*[Actual DAQ run number; unique scan ID]*
Campaign directory	*[Absolute persistent path]*
Gateware profile / identity	*[Active profile; live build ID; source commit]*
Software / environment	*[Server; PDS; decoder; DAQ release; local changes]*
Evidence manifest	*[Path to SHA-256 manifest and source/dirty-state record]*

2 Physical and logical configuration

[List management MAC/IP and actual data MAC/IP values separately from registered reservations. State any address mismatch and the DAQ configuration used.]

Setting	Recorded value
Links / senders / channels	*[Physical links; wire stream IDs; DAQ source IDs; selected channels]*
AFE mapping	*[Firmware AFE to physical board AFE mapping]*
DAPHNE endpoint / master	*[Address; master name; partition if relevant; state; locks; timestamp validity]*
SSP endpoint / master	*[Control address; timing address; master; state; readiness]*
Source configuration	*[For each master: generator, command ID, requested and measured rate, period, clock]*
DAQ window / trigger rate	*[Readout window in seconds and DAQ request rate in Hz]*
Optical/timing relationship	*[Optically coupled? Shared master? How verified?]*

3 Scan settings and method

Control	Recorded value and native units
Offset / VGAIN	*[DAC offset; VGAIN points; common to all AFEs or sequential?]*
Detector bias	*[DAPHNE detector-bias setting; separate from SSP LED bias]*
SSP LED selection	*[Controller channel count; mask; enabled channels]*
SSP LED bias	*[270 nm and 367 nm native codes]*
SSP pulse settings	*[Single/double mode; widths in ticks; delay in ticks; burst count]*
Per-point selection	*[Window duration; target count if used; settling; guard/tail]*
Final-state request	*[Paused or explicitly configured operating state]*

[Describe the per-point sequence: pause/drain, DAC writes, SSP configuration/readback, timing starts, acquisition, counter snapshots, stops, and cleanup. Include exact launcher command and script/configuration hashes.]

4 Results and acceptance

Denominators. Selected analysis windows contain *[waveforms per channel per point and total samples]*. Hardware counters cover *[time interval and accepted master commands per channel]*. These are different observation intervals; report both.

Point / code	Frames/channel	Pedestal range ADC counts	Within-frame RMS ADC counts	Rail clipping zero / full scale
[point]	*[count]*	*[min–max]*	*[min–max]*	*[fractions]*
[point]	*[count]*	*[min–max]*	*[min–max]*	*[fractions]*
[point]	*[count]*	*[min–max]*	*[min–max]*	*[fractions]*

[Attach pedestal/RMS versus scan coordinate and representative waveform figures. Define pedestal, total RMS, within-frame RMS, and waveform-mean drift. State any rejection, baseline subtraction, or filtering.]

Check	Pass/fail/not tested, and evidence
Expected gateware identity and runtime health	*[Observed identity and status log]*
Endpoint clock lock and timestamp readiness	*[Before/after snapshots for both devices]*
Every selected source/channel present	*[Source IDs, channel counts, receipt paths]*
Common DAPHNE event timestamps; no gaps/duplicates	*[Windows and audit result]*
Fragment status / format / stream mapping	*[Status bits and decoder checks]*
Hardware records versus accepted master commands	*[Per-channel increments and master denominator]*
Busy/full drops and rejected commands	*[Counts and observation interval]*
SSP bias/duration readbacks	*[Requested versus read-back values]*
SSP command counter response	*[Counter delta; SSP master; no cross-domain inference]*
Input hashes and official decoder cross-check	*[Manifest and decoder revision]*
ADC rail clipping and anomalous channels	*[Fractions; affected channels/points]*

5 Final state, limitations, and next action

[Record whether DAQ stop-run succeeded, whether applications are in error, DAPHNE pause mask, stopped timing sources, SSP LED-disable readback, and final endpoint readiness. Include cleanup failures explicitly.]

[State what was not established: optical output, detector response, physical gain, shared-master synchronization, reboot persistence, or other configuration limits. Describe the next test, owner, and required setup.]

6 Artifacts and review

- Raw HDF5 files, record IDs, windows, and hashes: *[paths/manifest]*.
- Selected waveform binaries and receipts: *[paths/hashes]*.
- Plan, runner, acquisition adapter, SSP driver and routine: *[versions/hashes/dirty state]*.
- Numeric analysis, plots, and decoder provenance: *[paths/hashes]*.
- Failed/interrupted attempts retained separately: *[IDs and disposition]*.
- Reviewer / date / status: *[not reviewed, reviewed, or approved; name the authority if approved]*.